

SILVER BAY, MN, USA

WMO: 727556

Lat: 47.249N

Lon: 91.416W

Elev: 332

StdP: 97.40

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04905

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.5	-25.9	-32.8	0.2	-27.6	-30.9	0.2	-24.3	8.1	-6.1	7.4	-6.7	0.9	290	0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	29.0	20.3	27.5	18.9	26.2	18.0	21.8	27.1	20.5	25.5	19.4	24.0	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.3	25.1	18.7	14.1	23.8	17.6	13.1	21.7	65.1	27.0	60.6	25.8	56.9	24.0	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.0	5.7	DB	-33.5	32.4	3.8	1.8	-36.3	33.7	-38.5	34.8	-40.6	35.8	-43.3	37.1	(4)
(5)			WB	-33.5	23.9	3.7	1.2	-36.2	24.8	-38.3	25.5	-40.4	26.1	-43.1	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-11.7	-10.1	-4.0	2.6	9.0	14.5	18.0	16.9	12.6	5.8	-1.7	-8.6
	DBStd	11.57	7.33	6.80	6.52	4.54	4.50	3.88	3.36	3.23	4.43	4.25	5.61	6.69
	HDD10.0	3076	674	562	434	226	73	8	0	1	24	145	352	577
	HDD18.3	5429	932	796	692	473	289	124	48	68	178	390	602	836
	CDD10.0	766	0	0	1	3	43	143	247	213	102	13	0	0
	CDD18.3	76	0	0	0	0	2	9	37	22	6	0	0	0
	CDH23.3	1074	0	0	0	1	50	162	493	287	78	3	0	0
	CDH26.7	241	0	0	0	0	8	31	129	59	13	0	0	0
Wind	WSAvg	2.1	2.1	2.2	2.3	2.7	2.6	2.0	1.6	1.6	1.7	2.2	2.1	2.0
Precipitation	PrecAvg	805	34	28	43	69	78	99	90	81	84	92	59	41
	PrecMax	946	77	81	89	239	141	183	198	147	168	203	127	94
	PrecMin	591	6	5	12	15	46	21	35	37	14	21	12	7
	PrecStd	107	18	19	19	50	26	40	49	32	35	47	36	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	7.7	15.1	22.1	28.0	29.6	32.2	31.0	28.6	22.7	14.8	6.3
		MCWB	1.9	3.2	8.2	11.7	17.3	21.2	21.5	21.5	20.1	15.1	9.9	2.6
	2%	DB	2.3	3.7	11.1	17.4	24.8	27.3	29.1	27.9	26.0	18.8	11.1	2.9
		MCWB	0.8	1.2	5.9	8.8	15.1	18.0	20.5	19.6	18.9	12.9	7.8	1.5
	5%	DB	0.9	2.1	7.6	13.8	21.3	25.2	27.8	26.4	22.9	16.1	8.0	2.2
		MCWB	-0.2	0.3	3.5	6.8	13.1	16.9	19.4	18.5	16.8	11.4	5.5	1.1
	10%	DB	-1.3	-0.1	5.0	11.3	17.9	22.8	26.3	24.7	21.1	12.9	6.3	1.0
		MCWB	-2.1	-1.5	1.7	5.6	11.5	15.9	18.4	17.8	16.2	9.4	4.1	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	4.0	10.3	13.0	19.5	22.4	23.9	22.9	21.8	16.5	11.2	3.6
		MCDB	4.4	6.6	13.4	19.0	24.8	28.3	29.2	28.1	26.3	20.6	13.7	5.2
	2%	WB	0.9	1.8	6.3	9.7	16.3	20.1	22.1	21.3	19.8	14.1	7.9	2.0
		MCDB	2.1	3.3	9.7	15.8	22.0	24.9	27.5	26.3	23.7	17.3	9.8	2.8
	5%	WB	-0.7	0.1	4.0	7.8	14.3	18.1	20.7	20.0	18.4	11.9	6.2	0.8
		MCDB	0.3	1.7	7.2	12.9	19.5	22.6	26.0	24.6	21.8	14.8	8.2	1.7
	10%	WB	-2.4	-1.7	2.0	5.8	12.4	16.8	19.5	18.9	16.8	10.1	4.1	-0.4
		MCDB	-1.8	-0.4	4.9	10.7	17.1	21.7	24.5	23.2	19.9	13.0	6.3	0.7
Mean Daily Temperature Range	5% DB	MDBR	11.1	12.5	12.2	11.6	12.8	12.9	13.3	13.1	12.7	10.2	9.0	9.4
		MCDBR	10.0	13.2	14.4	17.9	18.6	16.9	15.6	15.6	15.7	15.5	12.2	9.1
		MCWBR	8.4	10.5	9.8	10.7	10.1	9.1	7.9	8.6	9.5	10.1	8.8	7.6
	5% WB	MCDBR	9.5	11.2	13.2	16.2	15.2	14.0	14.3	13.6	13.6	13.1	11.0	8.0
		MCWBR	8.2	9.1	9.3	10.3	9.1	8.7	8.3	8.4	9.0	9.3	8.4	7.0
Clear-Sky Solar Irradiance	taub		0.253	0.278	0.310	0.342	0.368	0.378	0.388	0.379	0.358	0.313	0.289	0.266
	taud		2.440	2.366	2.374	2.385	2.385	2.390	2.395	2.437	2.466	2.560	2.526	2.455
	Ebn at Noon		857	897	915	912	896	886	871	866	853	851	805	793
	Edh at Noon		65	87	101	111	115	116	114	105	93	72	61	58
All-Sky Solar Radiation	RadAvg		1.27	2.24	3.47	4.62	5.39	5.80	6.10	5.14	3.81	2.30	1.28	0.95
	RadStd		0.14	0.23	0.32	0.48	0.38	0.40	0.25	0.34	0.23	0.26	0.16	0.10

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page